

INNOVA JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATION
in preparation for General Certificate of Education Advanced Level
Higher 2

CHEMISTRY

9729/03

Paper 3 Free Response

13 September 2017

2 hours

Candidates answer on separate paper.

Additional Materials: Writing Papers
Data Booklet
Cover Page

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions.

Section B

Answer **one** question.

You are advised to show all working in calculations.
You are reminded of the need for good English and clear presentation in your answers.
You are reminded of the need for good handwriting.
Your final answers should be in 3 significant figures.

At the end of the examination, fasten all your work securely together.
The number of marks is given in the brackets [] at the end of each question or part question.

This document consists of **12** printed pages.



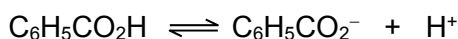
Section A

Answer **all** the questions in this section.

- 1 Benzoic acid, $\text{C}_6\text{H}_5\text{COOH}$ is a colourless crystalline solid and a simple aromatic carboxylic acid. Benzoic acid occurs naturally in many plants. Salts of benzoic acid are used as food preservatives and benzoic acid is an important precursor for the industrial synthesis of many other organic substances.

- (a) In the identification of benzoic acid, a commonly used reagent is neutral iron(III) chloride solution, in which a buff precipitate of iron(III) benzoate, $\text{Fe}(\text{C}_6\text{H}_5\text{CO}_2)_3$ is formed. When 50.0 cm^3 of iron(III) chloride solution was added to 50.0 cm^3 of benzoic acid solution, 0.0532 g of the buff precipitate was formed in the mixture.

The reactions that take place are shown below:



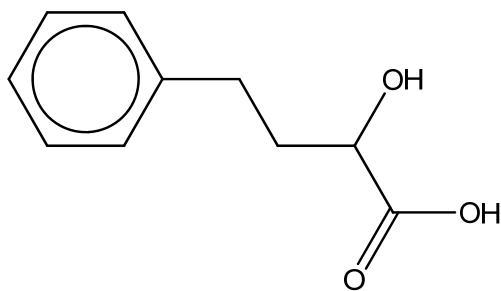
(Given: M_r of $\text{Fe}(\text{C}_6\text{H}_5\text{CO}_2)_3 = 418.8$)

- (i) Calculate the number of moles of benzoic acid that reacted with the neutral iron(III) chloride solution. [1]
- (ii) Given that the pH of the solution after the reaction is 2.33, calculate the number of moles of H^+ in the mixture. [1]
- (iii) Assuming that the H^+ ions in solution are formed only from the dissociation of benzoic acid in solution, as well as the reaction of benzoic acid with iron(III) chloride, calculate the concentration of benzoate ions in the mixture. [2]
- (iv) Iron(III) benzoate is a sparingly soluble salt.

Calculate the K_{sp} of iron(III) benzoate, given that the concentration of Fe^{3+} in the mixture was $1.83 \times 10^{-3} \text{ mol dm}^{-3}$ at equilibrium.

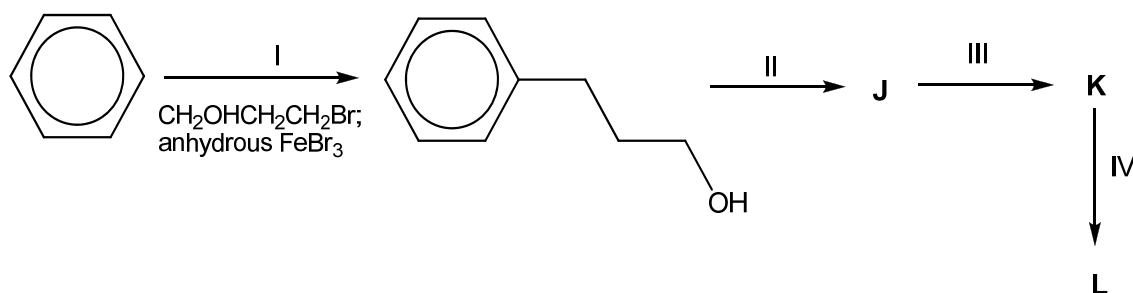
[1]

- (b) Benzoic acid can be produced by oxidising compound L.



Compound L

Compound L can be synthesised by the following reaction scheme.

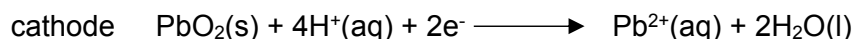
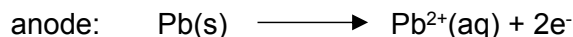


- (i) State the *type of reaction* in step I and hence explain the need for FeBr_3 to be anhydrous. [2]
- (ii) Suggest the identities of J and K. [2]
- (iii) State the reagents and conditions for steps II to IV. [3]
- (iv) State the *type of reaction* for stage IV. [1]

[Total: 13]

- 2 Sulfur is a chemical element with the symbol S and an atomic number of 16. It is an abundant, multivalent non-metal. It can be found in amino acids and it is a precursor to other chemicals such as H_2SO_4 . H_2SO_4 is a common mineral acid with many uses. It can be used as an electrolyte in batteries.

(a) In the cells of a lead-acid car battery the following reactions take place.



(i) Use the *Data Booklet* to calculate $E^\ominus_{\text{cell}}$ for this reaction. [1]

(ii) Construct an equation for the overall reaction. [1]

The electrolyte in a lead-acid cell is $\text{H}_2\text{SO}_4(\text{aq})$. Most of the $\text{Pb}^{2+}(\text{aq})$ ions that are produced at the electrodes are precipitated as a highly insoluble $\text{PbSO}_4(\text{s})$.

(iii) Construct an equation for the overall cell reaction in the presence of H_2SO_4 . [1]

(iv) By considering the effect of decreasing $[\text{Pb}^{2+}(\text{aq})]$ on the electrode potentials of the cathode and the anode, state and explain whether the overall $E^\ominus_{\text{cell}}$ will *increase, decrease or remain the same*. [1]

(b) H_2SO_4 is also used as an electrolyte in the anodising of aluminium. Anodising of aluminium is the process of coating aluminium metal with aluminium oxide (Al_2O_3) via electrolysis with the aluminium metal as the anode.

(i) Using $\text{H}_2\text{SO}_4(\text{aq})$ as the electrolyte and an inert electrode, draw an electrolysis set-up to show how a piece of aluminium metal can be anodised. [1]

(ii) Write chemical equations to show the reactions at the anode and cathode during anodising. Include in your answers, the overall equation. [3]

(iii) The aluminum piece to be anodised has a surface area of 29.2 cm^2 . Calculate the time taken to form a 0.2 mm protective layer of Al_2O_3 on the aluminum piece if a current of 2.0 A is passed through the set-up.

(Density of Al_2O_3 is 3.95 g cm^{-3}) [4]

(iv) Give one example of an anodised aluminium object, and explain the advantages of anodising it. [1]

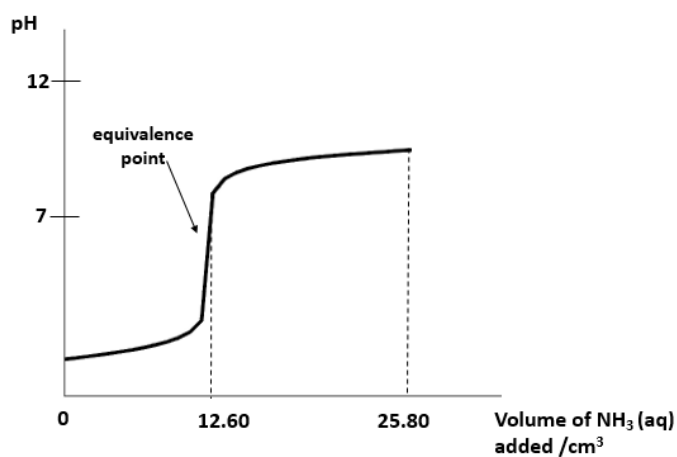
- (c) H_2SO_4 can be produced when sulfur trioxide is added to water, according to the following equation:



Calculate the volume of gaseous SO_3 , under room temperature and pressure, needed to form 30.0 cm^3 of $0.0200 \text{ mol dm}^{-3}$ of H_2SO_4 .

[1]

- (d) 25.0 cm^3 of $0.0200 \text{ mol dm}^{-3}$ of H_2SO_4 was titrated against $\text{NH}_3(\text{aq})$. The following graph was obtained.



- (i) Calculate the initial pH of the sulfuric acid solution.
- (ii) Calculate the concentration of $\text{NH}_3(\text{aq})$ used in this titration.
- (iii) Suggest a suitable indicator for this titration, giving a reason for your choice.

[1]

[2]

[2]

- (e) Thiophenol are the sulfur analogue of phenol, that is, sulfur takes the place of oxygen in the hydroxyl group of phenol. A common reaction that phenol and thiophenol have is shown in **Figure 1** below.

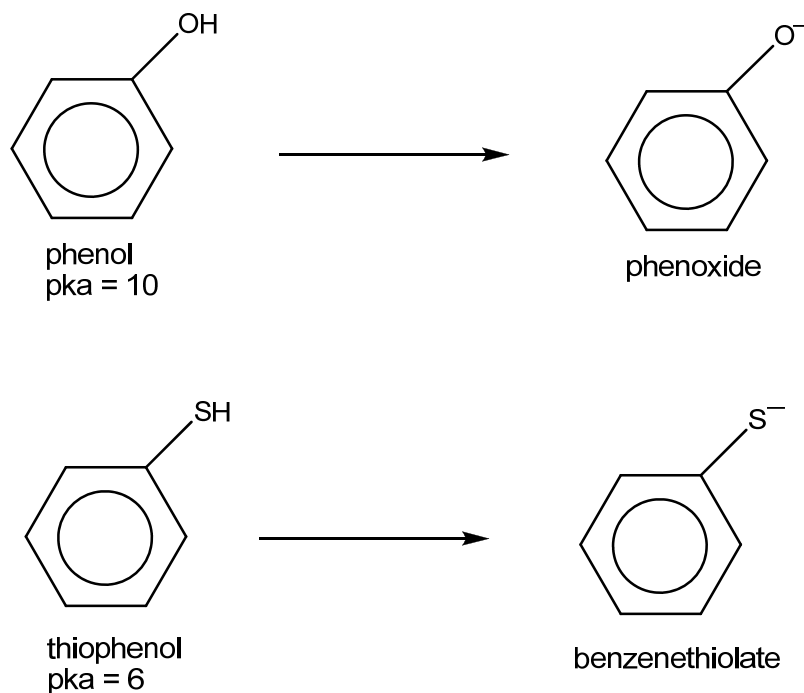
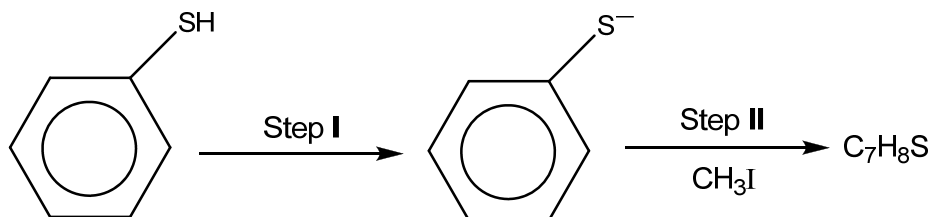


Figure 1

Consider the reaction scheme below.



- (i) Suggest reagent and condition for step I. [1]
- (ii) What *type of reaction* is step I and step II? [2]
- (iii) Suggest the structure of $\text{C}_7\text{H}_8\text{S}$. [1]

[Total: 23]

- 3 Halogenated organic compounds are widespread throughout nature and have a vast array of uses in modern industrial processes. They find many uses in the industries such as solvents and pesticides.

The rate of hydrolysis of $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{Br}$ with aqueous NaOH was studied in a series of 3 experiments in which the initial rate of the reaction was measured. The following results were obtained.

Experiment	$[\text{OH}^-] / \text{mol dm}^{-3}$	$[\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{Br}] / \text{mol dm}^{-3}$	Initial Rate / $\text{mol dm}^{-3} \text{ s}^{-1}$
1	0.100	0.010	4×10^{-6}
2	0.200	0.010	8×10^{-6}
3	0.200	0.020	16×10^{-6}

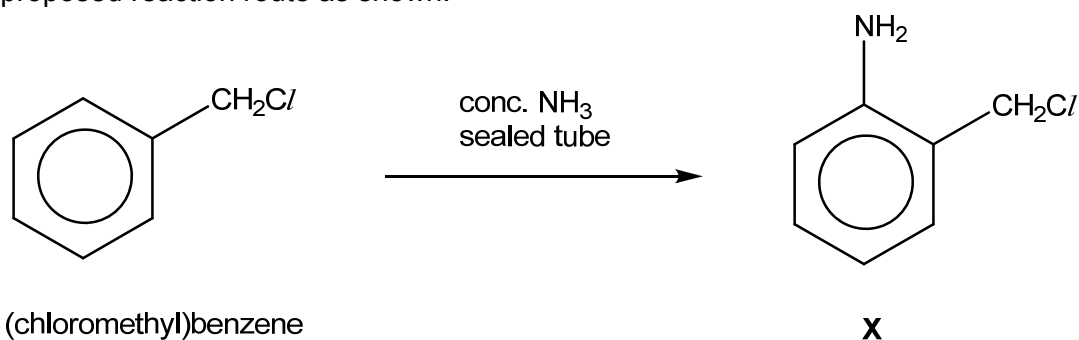
(a) Explain the meaning of the following terms.

- (i) order of reaction [1]
- (ii) half-life [1]
- (iii) Using the data given, derive the rate equation for the hydrolysis of $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{Br}$. [3]
- (iv) Using your rate equation, determine a value for the rate constant, including units for this reaction. [2]
- (v) Using the Maxwell-Boltzmann distribution curve, explain how the reaction rate might change with an increase in temperature. [4]
- (vi) Given that the rate equation for the hydrolysis of $(\text{CH}_3)_3\text{CBr}$ is $\text{rate} = k[(\text{CH}_3)_3\text{CBr}]$, suggest possible mechanisms for **both** of the reactions of $(\text{CH}_3)_3\text{CBr}$ and $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{Br}$ with aqueous NaOH which are consistent with the observed kinetics. [6]

(b) Bromine reacts with 2-methylpropane in the presence of sunlight to produce mono-brominated products via a free radical mechanism.

- (i) Explain why the mono-bromination of 2-methylpropane results in the formation of two products in unequal amounts. [1]
- (ii) By quoting appropriate data from the *Data Booklet*, explain the difference in the reactivity of fluorine with 2-methylpropane compare with that of bromine. [2]

- (c) A student was not successful in making **X** from (chloromethyl)benzene with the proposed reaction route as shown.



Instead, a different product with molecular formula $\text{C}_7\text{H}_9\text{N}$ was obtained.

- (i) Draw the displayed formula of the product obtained. [1]
- (ii) Suggest how **X** can be synthesised from (chloromethyl)benzene. You should include the reagents, conditions and intermediate(s) formed in your answer. [3]

[Total: 24]

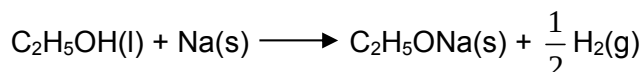
Section B

Answer **one** question from this section

- 4 Hydrogen gas is needed in a large number of processes in the chemical industries such as hydrocracking of petroleum, production of margarine and production of ammonia.

Hydrogen gas can be produced in many chemical reactions.

- (a) One of the reactions that produces hydrogen is the reaction of sodium with ethanol.



This is also a chemical test for the presence of alcohol.

- (i) Write the full electronic configuration of Na. [1]

- (ii) Describe how you would use the ionisation energies found in the *Data Booklet* to conclude that sodium is in group 1 of the periodic table. [1]

- (iii) Ethanoic acid reacts with sodium in a similar way. However, when ethanol and ethanoic acid is reacted with solid sodium carbonate separately, only ethanoic acid gives off a gas that forms white precipitate with calcium hydroxide.

Explain the difference in reactions of ethanol and ethanoic acid with sodium carbonate.

[3]

- (iv) An unknown compound **M**, $\text{C}_8\text{H}_{10}\text{O}$ was found to be an alcohol when tested with sodium metal.

When **M** is treated with hot acidified potassium dichromate, compound **N**, $\text{C}_8\text{H}_8\text{O}$ is formed. **N** gives a yellow precipitate, **P** when warmed with alkaline aqueous iodine.

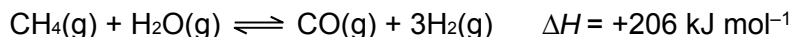
When **M** is heated with concentrated H_2SO_4 , compound **R**, C_8H_8 is formed. **R** gives compound **S** when reacted with hydrogen chloride gas.

M also gave compound **S** when reacted with phosphorous pentachloride at room temperature.

Deduce the structures of **M**, **N**, **P**, **R** and **S**.

[5]

- (b) Another reaction that produces hydrogen is the reaction of methane with steam at 1000–1400 K.



This reaction produces large quantities of hydrogen for industrial use.

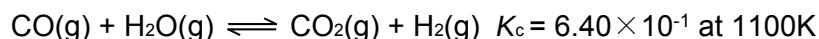
- (i) State and explain the effect of decreasing the temperature of the reaction on the equilibrium position. [1]
- (ii) Deduce the sign of entropy change for the forward reaction. [1]
- (iii) Hence, or otherwise, predict and explain the spontaneity of the forward reaction at low temperature. [1]
- (iv) Comment and explain if there is any discrepancy in your answers in **b(i)** and **b(iii)**. [1]
- (v) Calculate the enthalpy change of vaporisation of water using the data in **Table 1**.

Table 1

Enthalpy change of formation of carbon monoxide	–111 kJ mol ^{–1}
Enthalpy change of formation of methane	–75 kJ mol ^{–1}
Enthalpy change of formation of water	–285 kJ mol ^{–1}

[2]

- (c) Hydrogen gas can also be produced from the reaction of carbon monoxide with steam.

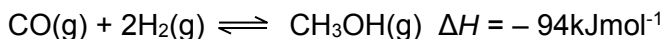


A mixture containing 0.8 mol of CO and 0.8 mol of H₂O was placed in a 2 dm³ flask and allowed to come to equilibrium at 1100K.

- (i) Write an expression for an expression for K_c of this reaction. [1]
- (ii) Calculate the amount of each substance present in the equilibrium mixture at 1100K. [2]
- (iii) State and explain the effect of decreasing pressure on the equilibrium constant. [1]

[Total: 20]

- 5 Methanol can be prepared industrially by reacting together carbon monoxide and hydrogen. This is a reversible reaction:



A chemist mixes together 0.114 mol CO(g) and 0.152 mol H₂(g) in a container. The container is pressurised and then sealed and the total volume is 200 cm³. The mixture is heated to 500 K at constant volume and left to reach equilibrium. The chemist analyses the equilibrium mixture and finds that 0.052 mol CH₃OH has formed.

- (a) Calculate the value of K_c for the equilibrium at 500 K. [2]
- (b) The chemist repeats the experiment using the same initial amounts of CO and H₂. The same procedure is used but the mixture is heated in the 200 cm³ sealed container to a temperature higher than 500 K.
As the gas volume is kept at 200 cm³, the increased temperature also increases the pressure.

Explain why it is difficult to predict how the yield of CH₃OH would change and state what happens to the value of K_c . [2]

- (c) Methanol can be oxidised to formaldehyde, HCHO, by passing its vapour over copper heated to 300 °C.

Explain why the boiling point of methanol (65 °C) is higher than that of formaldehyde (−19 °C). [2]

- (d) State the type of hybridisation shown by the C atom in formaldehyde and draw its hybrid orbitals. [2]

- (e) Benzaldehyde, C₆H₅CHO, is the simplest aromatic aldehyde and has a characteristic smell of almonds.

In the presence of a solution of potassium hydroxide, benzaldehyde undergoes disproportionation to give a mixture of two organic products, **E** and **F**. **F** has a higher solubility in water compared to **E**.

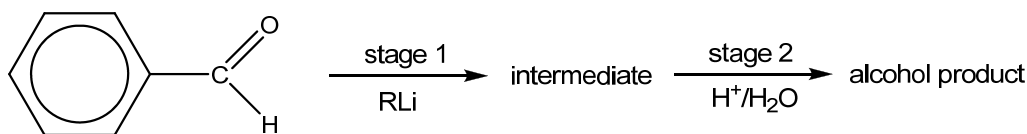
- (i) Suggest the structures of **E** and **F**. [2]

- (ii) Explain why **F** is soluble in water. [1]

- (f) Alkyl lithium compounds, RLi, can be used to increase the number of carbon atoms in an organic compound. Different alkyl groups, R, add carbon chains with different chain lengths.

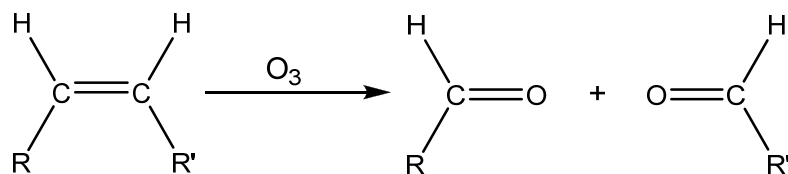
RLi provides a source of R[−] ions, which act as a nucleophile.

- (i) RLi reacts with benzaldehyde to give an intermediate in stage 1, followed by reaction with aqueous acid to form an alcohol in stage 2.



Describe the mechanism for the above reaction, including curly arrows and relevant dipoles. [3]

- (ii) 'Ozonolysis' is a technique used in organic chemistry to break open a C=C double bond. During ozonolysis, an alkene reacts with ozone, O₃. The products are carbonyl compounds as shown below.



An aromatic alkene, **A** undergoes ozonolysis to give benzaldehyde and **B**, C₄H₈O. **B** has no reaction with [Ag(NH₃)₂]⁺(aq) but gives a yellow precipitate with aqueous alkaline iodine.

B undergoes a reaction with an alkyllithium compound, **C**, followed by reaction with aqueous acid to give an alcohol, **D**, which is optically active. The elimination of H₂O from **D** produces a mixture of four different isomeric alkenes with the formula, C₇H₁₄, only two of which are cis-trans isomers of each other. Suggest the structural formulae of compounds **A** to **D** and the two cis-trans isomers.

[6]

[Total: 20]